CSC 5003 Algorithm Art and Programming Practice
Lecture 6: Search and Backtracking

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Word Search

- Given a m*n grid, check if the given word can be concatenated by neighbors.

- DFS + Backtracking

- How to do this backtracking?

- What's the worst case complexity?

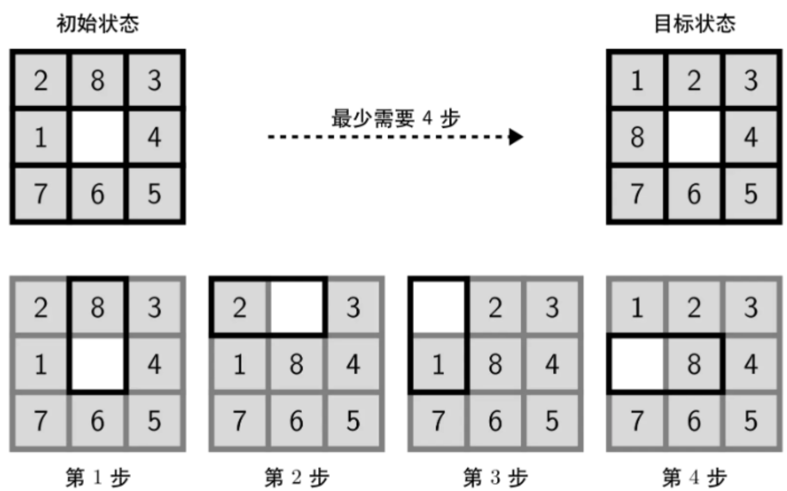
N-Queen

- Find ways of putting N queens in the chessboard.

- DFS + Backtracking

Eight-Digit

- Find the minimum step to reach the final state.



- BFS + HashMap

Stamping The Sequence

- You are given two strings stamp and target. Initially, there is a string s of the same length as target filled with question marks '?'. You can perform a stamping operation as follows:
  - Choose an index i such that 0 <= i <= target.length - stamp.length.
  - For every j from 0 to stamp.length - 1, set s[i + j] = stamp[j].
  - A position in s may be overwritten multiple times during the process.
- Your goal is check if the string target from the initial string of all '?' by performing stamping operations.

- Is this a search problem???

- Hint: Do it reversely.

Yet another "search" problem

- You are given a list of strings words from an alien language. The strings in words are **sorted lexicographically** according to the rules of this alien language.
- Your task is to **derive a possible order of characters** in this alien language and return it as a string.

- Hint: Model it into a graph.

Mid-term Review

- STL
  - Vector, map, set are most common.
- Trees and Graphs
  - DFS tree to find cycles.
- Topsort and Shortest Path
  - Focus on how to model into graphs.
- String Processing
  - Hashing is most important.

3/15 1:30 - 4:30

- At least 2 problems of Midterm are selected from the candidate sets.
- At most 1 problem from today.
- 1 problem not from class
- 4 problems in total.